

Lab Goal : This lab was designed to teach you more about a linked list.

Lab Description : Write a program that demonstrates that your BasicList works as intended. BasicList is a single-linked linked list. Each node in BasicList has a data component and a reference to the next node only. Write and test all methods in BasicList. Complete methods to add, remove, get, and search.

PART 1 – Open the BasicList.java file and complete the BasicList class.

PART 2 – Use lab15f.java to test your BasicList class.

ListNode – stores a value and a reference to the next node

```
public class ListNode implements Linkable
{
    private Comparable listNodeValue;
    private ListNode nextListNode;

    public ListNode(){
        listNodeValue = null;
        nextListNode = null;
    }

    public ListNode(Comparable value, ListNode next){
        listNodeValue=value;
        nextListNode=next;
    }

    public Comparable getValue(){
        return listNodeValue;
    }

    public ListNode getNext(){
        return nextListNode;
    }

    public void setValue(Comparable value){
        listNodeValue = value;
    }

    public void setNext(Linkable next){
        nextListNode = (ListNode)next;
    }
}
```

Sample Data :

```
4 5 6 7 8 9 10 11 12 13
24 75 86 37 82 94 111 82 43
0 4 5 2 1 4 6
```

Sample Output :

```
SUM:: 85.0
AVERAGE:: 8.5
SMALLEST:: 4
LARGEST:: 13
```

```
SUM:: 634.0
AVERAGE:: 70.44444444444444
SMALLEST:: 24
LARGEST:: 111
```

```
SUM:: 22.0
AVERAGE:: 3.142857142857143
SMALLEST:: 0
LARGEST:: 6
```